

DNA-Based Prenatal and Pre-Implantation Genetic Testing

Abbreviations:

EB, epidermolysis bullosa; IFM, immunofluorescence mapping; NGS, next-generation sequencing; TEM, transmission electron microscopy.

- **Immunofluorescence mapping (IFM)** or immunohistochemistry enables rapid diagnosis, which is critical for determining prognosis in newborns. Severe subtypes of EB—such as severe junctional EB (JEB) or severe recessive dystrophic EB (RDEB)—can be identified based on the level of skin cleavage and the absence or marked reduction of specific proteins.
- In cases with mild skin fragility, IFM and TEM may not provide a definitive diagnosis.
- **Transmission electron microscopy (TEM)** can identify the precise level of skin cleavage and structural abnormalities in keratin intermediate filaments, hemidesmosomes, basement membrane, or anchoring fibrils. It remains useful for early diagnosis in certain EB subtypes, such as severe dominant epidermolysis bullosa simplex (EBS).

4. Delivery of Results

A clinician experienced in EB should communicate the final diagnosis, including the EB subtype, prognosis, and inheritance pattern. However, even with a confirmed genetic diagnosis, precise prognostic predictions—especially in neonates—may not always be possible.

It is essential to:

- Allow sufficient time for discussion.
- Involve a psychologist when needed.
- Provide a translator or cultural mediator, if required.

Diagnosing EB in neonates presents emotional challenges, particularly in conveying the lifelong, chronic nature of the disease. Explanations may need to be repeated. In severe or lethal forms (e.g., severe JEB), management decisions should consider the family's social, cultural, and emotional context.

Genetic counselling must always be provided.

To enable DNA-based prenatal diagnosis in future pregnancies, an EDTA blood sample from the affected infant (index case) should be collected promptly with parental consent—especially in severe, potentially lethal forms.

5. Prenatal Diagnosis

Once the familial pathogenic variant(s) are identified and parental carrier status is confirmed, **DNA-based prenatal testing** and **pre-implantation genetic testing** can be offered upon request.

Availability of these tests may vary due to local legal, ethical, and cultural regulations.

Part II. Common Clinical Concerns in EB: Practical Recommendations for Healthcare Providers

Wound Care in EB

Wound care is the cornerstone of EB management. In the absence of a cure, supportive wound care and early detection and treatment of complications are the mainstays of therapy.

1. Clinical and Laboratory Assessment

Effective wound care must take into account the following factors:

a. EB Subtype and Patient Age

The severity of mucocutaneous and extracutaneous manifestations, as well as wound characteristics, varies significantly among EB subtypes. These are influenced by:

- The causative gene
- Type of mutations
- Patient age

An initial assessment should always include:

- Estimation of body surface area affected
- Characterization of lesions (blisters, erosions, ulcers, hyperkeratosis)

b. Nutritional Status, Chronic Anaemia, and Hypoalbuminaemia

Malnutrition, chronic anaemia, and hypoalbuminaemia are common—particularly in severe EB subtypes such as JEB and RDEB.

Contributing factors include:

- Reduced nutrient intake
- Chronic inflammation
- Protein and nutrient loss through extensive, non-healing wounds

These conditions impair wound healing and must be actively managed.

- Regular nutritional assessment by a dietitian is essential.
- In children, monitor growth curves.
- Routinely check haemoglobin and serum albumin levels.